

VERIQO OS Architecture

Status: FROZEN — MIP-001 W0 **Supersedes:** nothing. Sits above every prior architecture document as the integrating frame; prior documents remain valid within their scope.

1. The reframing

VERIQO is **not** maritime AI plus dark web plus insurance plus dispute resolution plus an LLM. That composition produces what the architecture review correctly called a *Frankenstein platform* — fifteen products sharing a logo.

VERIQO is an **Evidence Operating System**. Maritime, commodity, supply chain, insurance, trade finance, eBL, financial investigation, dispute, and dark web are **domain intelligence capabilities that consume the same evidence fabric**.

The architectural law:

ONE CANONICAL STATE. MANY DOMAIN PROJECTIONS.

The second law:

INTELLIGENCE MAY PROPOSE. EVIDENCE AUTHORITY MUST REMAIN DETERMINISTIC, POLICY-BOUND, AUDITABLE, REPLAYABLE, AND HUMAN-GOVERNED WHERE REQUIRED.

The third law:

NO DOMAIN MAY BYPASS THE EVIDENCE, POLICY, RIGHTS, PROVENANCE, QUALIFICATION, DISCLOSURE, OR REPLAY FABRIC.

2. Five planes

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| 1. INTELLIGENCE PLANE
Aureum · God of EYS · LLM · Dark Web · ML · Analytics
May PROPOSE. Holds no evidence authority. |
| 2. EPISTEMIC PLANE
Reverse Proof · Hypothesis · Contradiction · Trust
Observability · Qualification · Next Best Evidence |
| 3. EVIDENCE PLANE
IEAP · Raw · Provenance · Lineage · WORM
Independence · Integrity · Custody · Versioning |
| 4. GOVERNANCE PLANE
NEP · Rights · Privilege · PO · Disclosure · Dissent
Conflict · Procedural Neutrality · AI Authority |
| 5. TRUST / EXECUTION KERNEL
Identity · Policy · Determinism · Ledger · Raft
Replay · Cryptography · KMS/HSM · Attestation |

The invariant that makes this an OS rather than a diagram: a domain sits only at plane 1 and reaches planes 2–5 through their published contracts. There is no path from a domain adapter to storage, ledger, or policy that skips the intervening planes. pkg/constitution's Article 8 and the third law above are the executable form of this.

3. Canonical evidence lifecycle

Every case follows this, including dark web. There is not a separate "fast path".

CLAIM / QUESTION → CASE CONSTITUTION → REVERSE PROOF

→ WHAT MUST BE TRUE? → WHAT SHOULD WE OBSERVE?

→ NEXT BEST EVIDENCE → RIGHTS + AUTHORITY

→ INDEPENDENT ACQUISITION → RAW PRESERVATION

→ INTEGRITY + PROVENANCE → SOURCE INDEPENDENCE

→ OBSERVABILITY GATE → CONTRADICTION MATRIX

→ ALTERNATIVE HYPOTHESES → TRUST CALCULUS

→ DISSENT → EPISTEMIC QUALIFICATION

→ PROCEDURAL QUALIFICATION → DISCLOSURE GOVERNANCE

└─ ORIGINAL → WORM

└─ DERIVATIVE → REDACTION / ZKP

→ COMMON FACT PACK → INDEPENDENT VERIFIER → REPLAY

Note the ordering that is easy to get wrong and expensive to fix: **rights precede acquisition, raw preservation precedes integrity assessment, and observability precedes contradiction** — because you cannot score a contradiction against evidence you were never in a position to observe.

4. Where each plane lives in this repository

The architecture is not a greenfield. This table is the binding map from the plane model onto code that already exists, and it is what MIP §7 ("does this capability already exist?") is answered against.

Plane	Capability	Existing implementation	Status
5	Ledger	pkg/platform/audit (hash-chained, Merkle)	EXISTS
5	Policy / purpose binding	pkg/authz	EXISTS
5	Identity	pkg/identity, pkg/platform/security	EXISTS
5	Determinism / canonicalization	pkg/canonical/jcs	EXISTS
5	Consensus	pkg/consensus/raftlite, pkg/cluster	EXISTS
5	Crypto / KMS abstraction	pkg/platform/security/keys	EXISTS
5	Storage / WAL	pkg/storage, pkg/storage/wal	EXISTS
5	Replay	pkg/replay, commercialapi.Store.Replay	EXISTS
3	Evidence manifest + custody FSM	pkg/evidence/manifest (FROZEN)	EXISTS
3	Canonical evidence projection	pkg/commercial/evidencefabric	EXISTS
3	Retention / legal hold	pkg/governance/data	EXISTS
3	Provenance	pkg/evidence/provenance, pkg/insurance/cre/provenance.go	EXISTS
3	Acquisition contracts	pkg/connector/{aisstream, bol, insurance, payment, sar}	EXISTS (simulated sources)
2	Trust calculus	pkg/core/trustcalc, pkg/trust	EXISTS
2	Contradiction	pkg/insurance/contradiction, pkg/moat/contradiction	EXISTS
2	Hypothesis	pkg/insurance/causation	EXISTS
2	Dependency / corroboration	pkg/insurance/evidence/{dependency, corroboration}.go	EXISTS

Plane	Capability	Existing implementation	Status
2	Reverse proof	—	BUILT THIS ROUND (pkg/qualification/reverseproof)
2	Observed absence	—	BUILT THIS ROUND (pkg/qualification/observability)
2	Source independence (canonical)	—	BUILT THIS ROUND (pkg/qualification/independence)
2	Qualification states	—	BUILT THIS ROUND (pkg/qualification/state)
2	Next best evidence	—	BUILT THIS ROUND (pkg/qualification/nextbest)
4	Constitution (executable)	—	BUILT THIS ROUND (pkg/constitution)
4	Disclosure P/C model	—	BUILT THIS ROUND (pkg/disclosure/access)
4	Dissent	partial (contradiction tests)	CANONICALIZED THIS ROUND
1	AI evidence gateway	—	BUILT THIS ROUND (pkg/ai/gateway)
5	Canonical EventEnvelope	—	BUILT THIS ROUND (pkg/contract/event)
1	Aureum / God of EYS / dark web	—	NOT BUILT — see §6

5. The anti-duplication rule

MIP §7 forbids a second authoritative engine for a capability an existing primitive can carry. Concretely:

Forbidden:

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ledger + audit-ledger + disclosure-ledger + privilege-ledger + redaction-ledger
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Required:

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ONE ledger, carrying disclosure · redaction · privilege · dissent events
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Same for policy:

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ONE policy plane
├─ Acquisition  ── Rights      ── Disclosure
├─ AI           ── Privilege   ── Protective Order
├─ Neutrality   ── Qualification
```

This round adds **no new ledger and no new policy engine**. The new packages emit into the existing pkg/platform/audit chain and express their rules as data the existing policy plane evaluates.

6. What is deliberately not built

Named here rather than discovered later, per MIP §32 rule 18.

Capability	Why not built
Aureum AI, God of EYS	Require an actual model deployment and an investigation corpus. The gateway they must pass through is built and enforced; the intelligences themselves are not.

Capability	Why not built
Dark web acquisition	Requires legal review of acquisition authority per jurisdiction. The acquisition contract shape exists; no live acquisition path is built, and building one without that review would violate Article 4.
ZKP prover	A real proving system is a substantial cryptographic dependency. Article 9's <i>bounded meaning</i> rule is expressed in the constitution; no prover is built, and none is claimed.
PDF/XLSX/PPTX redaction workers	Require document-format toolchains and an adversarial recovery lab. The redaction state machine and assurance record are specified; the media workers are not built.
Multi-region, HSM tenancy, pentest	External procurement — see <code>docs/VERIQO_EXTERNAL_QUALIFICATION_TRACK.md</code> .

Per MIP §34, each of these is **DESIGNED**, not **IMPLEMENTED**. None is reported as done.

7. Commercial product modes

Private Intelligence Mode and Neutral Evidence Mode run on the **same** Evidence OS. The distinction is policy and configuration, never a separate architecture or a forked codebase. This is what makes the neutrality claim credible: a neutral case is not running different software that could quietly behave differently.

8. One-sentence lock

VERIQO is an Evidence Operating System that independently acquires, preserves, connects, tests, qualifies, bounds, discloses, verifies, and replays evidence — while Aureum, God of EYS, dark web, maritime, commodity, insurance, supply chain, finance, and dispute intelligence are capabilities wholly subject to the Evidence Constitution, IEAP, EQF, NEP, policy, ledger, cryptography, independent verification, and replay.